

Chapter 9

Communication Description

9.1 Modbus protocol

The parameters of the servo drive are classified into 16-bit and 32-bit parameters based on the data length. Data read/write for the parameters is implemented based on the Modbus RTU protocol. The command codes for reading/writing parameters vary with the data length.

Operation	Command Code
Read 16-bit/32-bit parameters	0x03
Write 16-bit parameters	0x06
Write 32-bit parameters	0x10

Read parameter command code: 0x03

Based on the Modbus RTU protocol, command code 0x03 is used to read both 16-bit and 32-bit parameters.

Request frame format:

Value	Description
START	Equal to or greater than 3.5-character idle time, frame start
ADDR	Servo axis addresses 1 to 247 Note: 1 to 247 are decimal values and need to be converted to hexadecimal equivalents.
CMD	Command code: 0x03
DATA[0]	Register start address (high 8 bits): parameter group number of the start register For example, for parameter C06.11, where 06 is the group number, that is, DATA[0] is 0x06.
DATA[1]	Register start address (low 8 bits): offset within the parameter group of the start register For example, for parameter C06.11, 11 is the offset within the parameter group, that is, DATA[1] is 0x11.
DATA[2]	High 8 bits of the number of parameters to be read (hexadecimal), N(H)
DATA[3]	Low 8 bits of the number of parameters to be read (hexadecimal), N(L)
CRCL	CRC active byte (low 8 bits)
CRCH	CRC active byte (high 8 bits)
END	Equal to or greater than 3.5-character idle time, frame end

Response frame format:

Value	Description
START	Equal to or greater than 3.5-character idle time, frame start
ADDR	Servo axis address, hexadecimal
CMD	Command code: 0x03
DALENGTH	Number of parameter bytes, equal to twice the number of parameters to be read, that is, $N \times 2$
DATA[0]	Data of the first register parameter (high 8 bits)
DATA[1]	Data of the first register parameter (low 8 bits)
DATA[...]	...
DATA[N*2 2]	Data of the Nth register parameter (high 8 bits)
DATA[N*2 1]	Data of the Nth register parameter (low 8 bits)
CRCL	CRC active byte (low 8 bits)
CRCH	CRC active byte (high 8 bits)
END	Equal to or greater than 3.5-character idle time, frame end

Based on the Modbus RTU protocol, command code 0x06 is used to write 16-bit parameters. Command code 0x10 is used to write 32-bit parameters.

Communication example

To read two-word data with C02.02 as the start register from the drive whose servo axis address is 01.

Master request frame:

01	03	02	02	00	02	CRCL	CRCH
----	----	----	----	----	----	------	------

Slave response frame:

01	03	04	00	01	00	00	CRCL	CRCH
----	----	----	----	----	----	----	------	------

The response frame indicates that the slave returns two-word (four-byte) data, the content of which is 0x0001 and 0x0000.

To read parameter C05.07 (32-bit) from the drive whose servo axis address is 01.

Master request frame:

01	03	05	07	00	02	CRCL	CRCH
----	----	----	----	----	----	------	------

Slave response frame:

01	03	04	00	01	00	00	CRCL	CRCH
----	----	----	----	----	----	----	------	------

The response frame indicates that the value of parameter C05.07 is 0x00000001.

Command code for writing 16-bit parameters: 0x06

 CAUTION

- Do not use 0x06 to write 32-bit parameters; otherwise, it will lead to serious errors.

Request frame format:

Value	Description
START	Equal to or greater than 3.5-character idle time, frame start
ADDR	Servo axis addresses 1 to 247 Note: 1 to 247 are decimal values and need to be converted to hexadecimal equivalents.
CMD	Command code: 0x06
DATA[0]	Register start address (high 8 bits): parameter group number of the start register For example, for parameter C06.11, where 06 is the group number, that is, DATA[0] is 0x06.
DATA[1]	Register start address (low 8 bits): offset within the parameter group of the start register For example, for parameter C06.11, 11 is the offset within the parameter group, that is, DATA[1] is 0x11.
DATA[2]	High 8 bits of data to be written to the register (hexadecimal)
DATA[3]	Low 8 bits of data to be written to the register (hexadecimal)
CRCL	CRC active byte (low 8 bits)
CRCH	CRC active byte (high 8 bits)
END	Equal to or greater than 3.5-character idle time, frame end

Response frame format:

Value	Description
START	Equal to or greater than 3.5-character idle time, frame start
ADDR	Servo axis address, hexadecimal
CMD	Command code: 0x06
DATA[0]	Register start address (high 8 bits): parameter group number of the start register For example, for parameter C06.11, where 06 is the group number, that is, DATA[0] is 0x06.

Value	Description
DATA[1]	Register start address (low 8 bits): offset within the parameter group of the start register For example, for parameter C06.11, 11 is the offset within the parameter group, that is, DATA[1] is 0x11.
DATA[2]	High 8 bits of data to be written to the register (hexadecimal)
DATA[3]	Low 8 bits of data to be written to the register (hexadecimal)
CRCL	CRC active byte (low 8 bits)
CRCH	CRC active byte (high 8 bits)
END	Equal to or greater than 3.5-character idle time, frame end

Communication example

To write data 0x0001 to parameter C02.02 in the drive whose servo axis address is 01.

Master request frame:

01	06	02	02	00	01	CRCL	CRCH
----	----	----	----	----	----	------	------

Slave response frame:

01	06	02	02	00	01	CRCL	CRCH
----	----	----	----	----	----	------	------

The response frame indicates that data 0x0001 has been written to parameter C02.02 in the drive whose servo axis address is 01.

Command code for writing 32-bit parameters: 0x10

CAUTION

- Do not use 0x10 to write 16-bit parameters; otherwise, it will lead to serious errors.

Request frame format:

Value	Description
START	Equal to or greater than 3.5-character idle time, frame start
ADDR	Servo axis addresses 1 to 247 Note: 1 to 247 are decimal values and need to be converted to hexadecimal equivalents.
CMD	Command code: 0x10
DATA[0]	Register start address (high 8 bits): parameter group number of the start register For example, for parameter C11.12, where 11 is the group number, that is, DATA[0] is 0x11.

Value	Description
DATA[1]	Register start address (low 8 bits): offset within the parameter group of the start register For example, for parameter C11.12, 12 is the offset within the parameter group, that is, DATA[1] is 0x12.
DATA[2]	High 8 bits of the number of parameters to which data will be written (hexadecimal), M(H) For example, to write data to parameter C03.02 alone, DATA[2] is 00, DATA[3] is 02, and M is 0002. One 32-bit parameter is counted as two words.
DATA[3]	Low 8 bits of the number of parameters to which data will be written (hexadecimal), M(L)
DATA[4]	Number of bytes corresponding to data to be written to the register, that is, M x 2 For example, to write data to parameter C03.02 alone, DATA[4] is 04.
DATA[5]	High 8 bits of data to be written to the start register (hexadecimal)
DATA[6]	Low 8 bits of data to be written to the start register (hexadecimal)
DATA[7]	High 8 bits of data to be written to Start register address + 1 (hexadecimal)
DATA[8]	Low 8 bits of data to be written to Start register address + 1 (hexadecimal)
CRCL	CRC active byte (low 8 bits)
CRCH	CRC active byte (high 8 bits)
END	Equal to or greater than 3.5-character idle time, frame end

Response frame format:

Value	Description
START	Equal to or greater than 3.5-character idle time, frame start
ADDR	Servo axis address, hexadecimal
CMD	Command code: 0x10
DATA[0]	Register start address (high 8 bits): offset within the parameter group of the start register For example, to write data to parameter C11.12, DATA[0] is Cx11.
DATA[1]	Register start address (low 8 bits): offset within the parameter group of the start register For example, to write data to parameter C11.12, DATA[1] is 0x12.
DATA[2]	High 8 bits of the number of parameters to which data will be written (hexadecimal), M(H)
DATA[3]	Low 8 bits of the number of parameters to which data will be written (hexadecimal), M(L)
CRCL	CRC active byte (low 8 bits)

Value	Description
CRCH	CRC active byte (high 8 bits)
END	Equal to or greater than 3.5-character idle time, frame end

Error response frame

Error frame response format:

Value	Description
START	Equal to or greater than 3.5-character idle time, frame start
ADDR	Servo axis address, hexadecimal
CMD	Command code: 0x90
DATA[0]~[3]	DATA error code
CRCL	CRC active byte (low 8 bits)
CRCH	CRC active byte (high 8 bits)
END	Equal to or greater than 3.5-character idle time, frame end

Error code:

Error code	Code description
0x0001	Illegal command code
0x0002	Illegal data address
0x0003	Illegal data
0x0004	Slave device fault
0x0006	Write length error
0x0008	Sector error
0x0020	Reading disabled

32-bit parameter addressing

When Modbus commands are used to read/write 32-bit parameters, the communication address is determined by the address of the parameter with the smaller offset number. Two parameters are operated per operation. See the following examples.

NOTICE

- In the following example, the servo axis address defaults to 01.

For example, the Modbus command for reading parameter C11.12 (1st displacement) is as follows:

01	03	11	12	00	02	CRCL	CRCH
----	----	----	----	----	----	------	------

If the "1st displacement" is 0x40000000 (decimal equivalent: 1073741824), the following response frames apply.

When COA.06 is set to 0 (Low 16 bits before high 16 bits), the response frame is as follows:

01	03	04	00	00	40	00	CRCL	CRCH
----	----	----	----	----	----	----	------	------

When COA.06 is set to 1 (High 16 bits before low 16 bits), the response frame is as follows:

01	03	04	40	00	00	00	CRCL	CRCH
----	----	----	----	----	----	----	------	------

For example, the Modbus command for writing 0x12345678 (1st displacement) is:

When COA.06 is set to 0 (Low 16 bits before high 16 bits), the response frame is as follows.

01	10	11	0C	00	02	04	56	78	12	34	CRCL	CRCH
----	----	----	----	----	----	----	----	----	----	----	------	------

When COA.06 is set to 1 (High 16 bits before low 16 bits), the response frame is as follows.

01	10	11	0C	00	02	04	12	34	56	78	CRCL	CRCH
----	----	----	----	----	----	----	----	----	----	----	------	------

For example, the command for writing 0x00100000 (decimal equivalent: 1048576) to the 32-bit parameter C05.07 is as follows.

If COA.06 is set to 0 (Low 16 bits before high 16 bits), the Modbus command is:

01	10	05	07	00	02	04	00	00	00	10	CRCL	CRCH
----	----	----	----	----	----	----	----	----	----	----	------	------

CRC

The host controller and the servo drive must use the same CRC algorithm during communication.

Otherwise, a CRC error will occur. The A6-RS servo drive adopts 16-bit CRC, with low bytes placed before high bytes. The polynomial used for CRC is $X_{16} + X_{15} + X_2 + 1$ (0xA001).

Uint16 COMM_CrcValueCalc(const Uint8*data, Uint16 length)

```
{
    Uint16 crcValue = 0xffff;
    int16 i;
    while (length--)
    {
        crcValue ^= *data++;
        for (i = 0; i < 8; i++)
        {
            if (crcValue & 0x0001)
```

```
{  
    crcValue = (crcValue>>1) ^ 0xA001;  
}  
else  
{  
    crcValue = crcValue>>1;  
}  
}  
}  
return (crcValue);  
}
```